# The SSD firmware bug that nearly killed our CephFS

## ...and how Ceph saved the day

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*This is a story about March 27th and the days that followed, when our main CephFS cluster suffered from SSDs going bad close together, how we diagnosed and resolved the outage.*
*And how Ceph saved our data.*

# Ceph at CERN

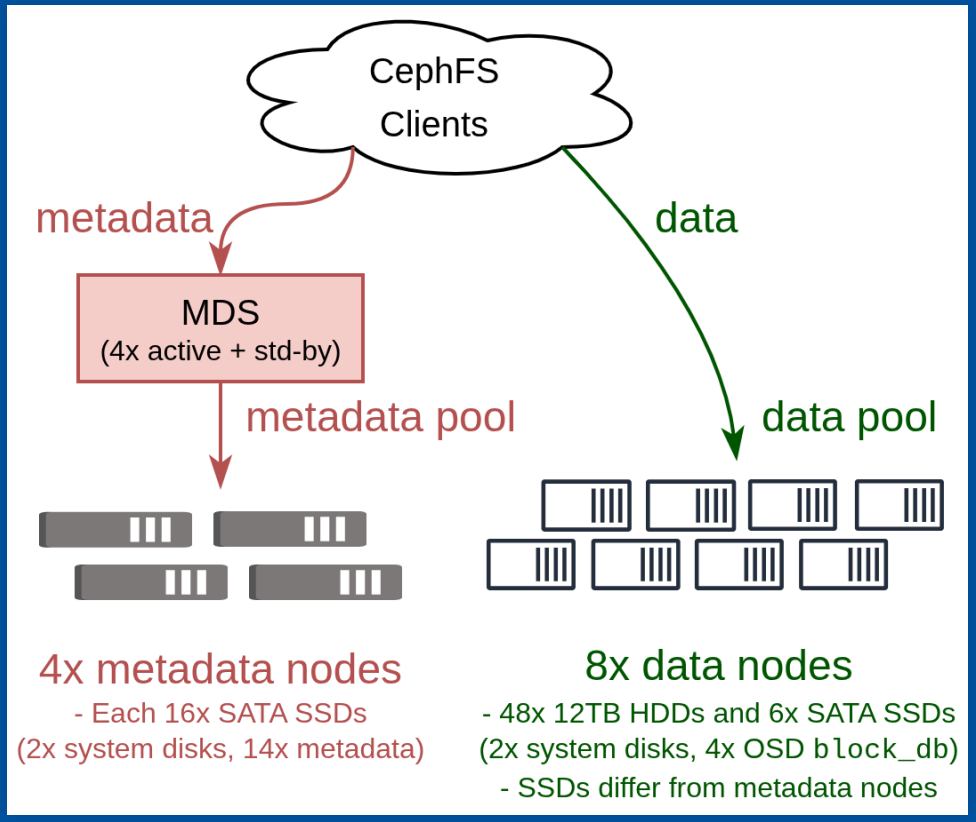
- Cornerstone of storage for IT Infrastructure since 2013
- 2 Data Centres, 22 clusters, 120 PB raw capacity
  - Blocks (OpenStack + `krbd`)
  - Objects (S3, Swift)
  - Files and Directories (kernel/fuse + NFS/SMB exports)

- 10x File Storage Clusters:
  - 6x HDD, 4x SSD
  - ➡️ CephFS HDD: Main, General Purpose CephFS
    - 2.2 PB used raw, 3200 clients, 300M files
- 9x Block Storage Clusters: 6x HDD, 3x SSD
- 3x Object Storage Clusters: All HDD

# “CephFS HDD”: What is it used for?

- Containerized applications (OKD)
  - Groupware, collaborative applications, issue tracking
  - Engineering applications
  - Web hosting
- WLCG Grid services
  - LHC@home
  - Job submission controllers
  - Test engines for clouds
- Repos for scientific publications and multimedia contents
- Monitoring, Software building, …

# “CephFS HDD”: What does it look like?



# Incident and Recovery

# Incident Timeline

- **Friday 27/03**
  - **00:35**: SSDs with metadata start to get sick
  - **06:45**: Investigations start
  - **13:50**: Root cause speculatively identified
  - **14:35**: First attemp to migrate metadata
  - **17:15**: Second attemp to migrate metadata
  - **18:50**: 75% of metadata moved
- **Saturday 28/03**
  - **02:43**: All metadata out of sick SSDs
  - **03:20**: MDS restart to clean up stuck operations
  - HEALTH_OK

# Detection

- **00:35**: Load climb (`io_wait`) on 1 metadata node
  (other 2 nodes will follow at 04:25 and 05:15)
- **00:45**: Various Ceph alarms go off:
  - SLOW_OPS
  - MDS_SLOW_REQUESTS
  - MDS_SLOW_METADATA_IO
  - MDS_CLIENT_LATE_RELEASE
- Other custom alarms go off:
  - MDS performance (e.g., `create/unlink latency`)
  - Errors accessing the FS and basic functional testing
- **06:45**: Wake up, look at phone 🔥

# Root Cause Analysis

- Check network, hardware, kernel errors, uptime, perf counters, wallclocks, user activity, ...
- Early results:
  - Extremely high IOwait and device utilization
  - No kernel logs about bad devices or slow IO
  - Devices suffer with Ceph OSD process running
    - Stopping OSD and dd-ing data out works ok

- Suspected HW, but random and widespread. Unlikely.
- Also, why do we get performance back with OSD stopped?
- Uncertainty: HW? Bug? Uptime? User-related? DDoS?

# Root Cause Analysis (cont'd)

- HW monitoring shows only metadata nodes are affected
- OSDs there are clearly complaining about slow devices:

```
log_latency_fn slow operation observed for _txc_committed_kv,
    latency = 8.204925537s, txc = 0x55d33c5d1880

_txc_state_proc slow aio_wait, txc = 0x55d33c5d1880
    latency = 5.988610744s
```

- Same problem elsewhere?

  **Yes:** Test cluster, same hardware, close-by rack.

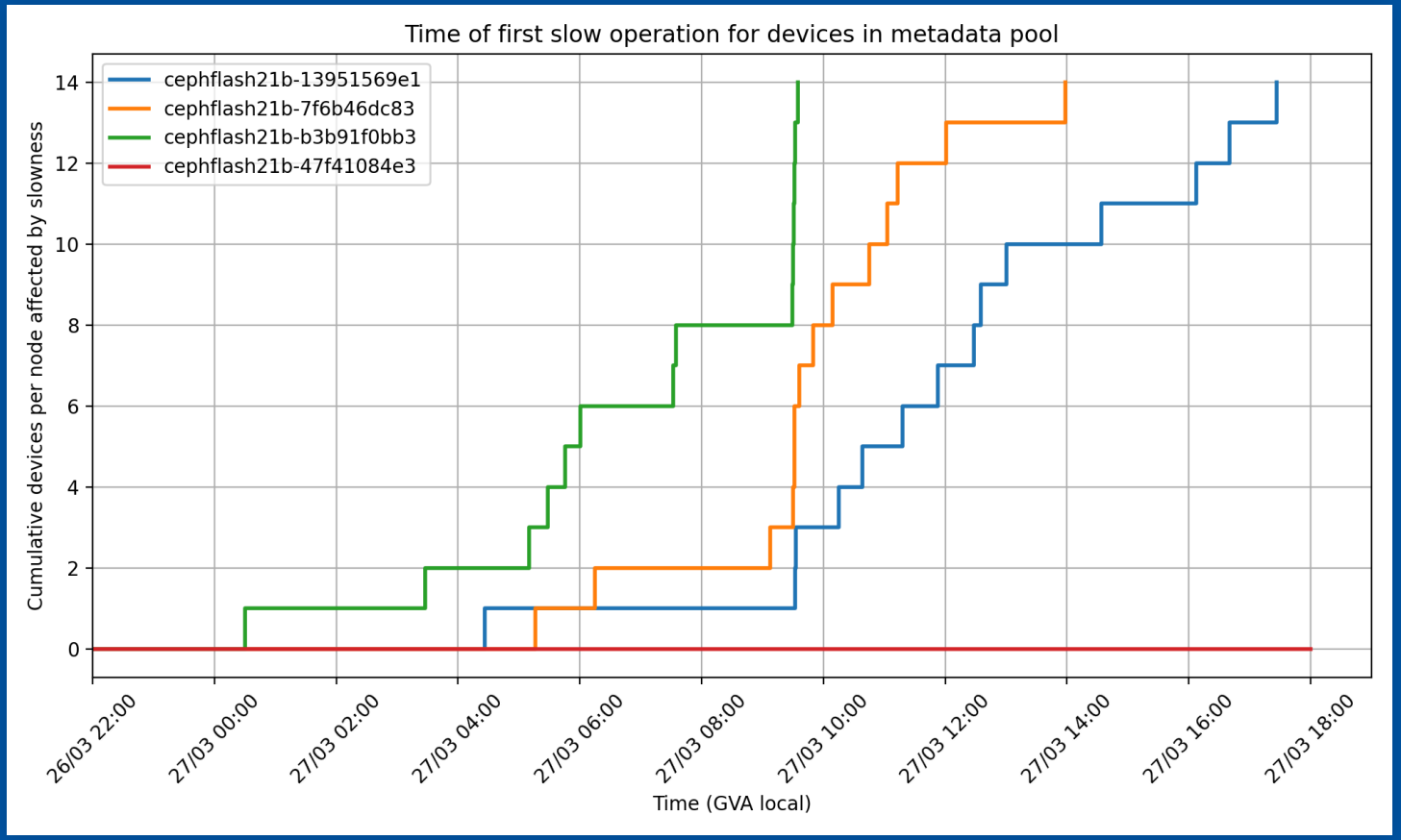
  - Reboot one node? Does not come back; **Avoid.**
  - Software stack? Much newer --> ~Not a Ceph bug.
  - Cluster load? Much lower --> Not user driven.

# Root Cause Analysis: The Realization

- It really looks like HW, but 1 node (of 4) still ok. Why?
  - SMART attribute `Power_On_Hours`
    - Bad nodes: 65'562 +/-12
    - Good node: 65'498
  - Suspiciously close to MAX `uint16`: 65'535

- **Best bet: Bug in SSD Firmware**
  - Action: Move the metadata out of sick SSDs.

# Slow Operations reported by OSDs



# Recovery: 1st attempt (14:30 to 17:00)

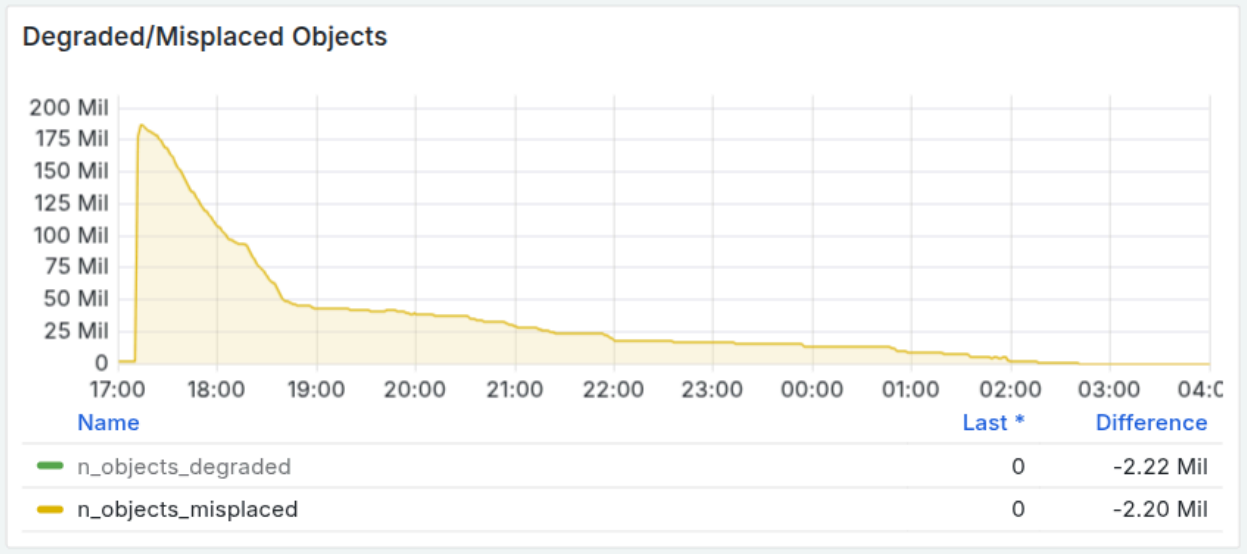
- Add new SSDs to replace sick devices:
  - Familiar dance of tuning CRUSH weights, draining, …
- It does not go well:
  - Load futher increases on sick SSDs
  - Tons of slow peering due to poor performance
  - Struggling to get data out efficiently
- 2hrs later, aborted and back to starting point:
  - Restored previous CRUSH settings
  - upmap-remapped all PGs to 0% misplaced

# Recovery: 2nd attempt (17:15 onward)

- Bounce metadata pool to HDDs used by data pool:
  - `ceph osd pool set <pool> crush_rule <cr>`
  - Metadata is stored as OMAPs, which live in `block.db`, and HDDs still have SSDs as DB device

- 75% of metadata moved in 1h30m:
  - ~Fast: We read the one replica from the healthy node
- Remaining 25% took ~8 hours:
  - Replicas on sick devices only
  - Heavy Ceph tuning to facilitate recovery

# Recovery: 2nd attempt (17:15 onward)



# Recovery: 2nd attempt (17:15 onward)

- All metadata out of sick SSD at 02:43, Saturday 28th
- Stuck peering requests among MDSs (`batch getattrs`)
  - Restarted authoritative MDS for that branch
- `HEALTH_OK`

# Side Effects

1. No meta/data loss or corruption
    - A few corrupted databases, recovered from backups
2. Negligible performance degradation from metadata move
    - Reminder: Metadata OMAPs are stored on `block.db`, which is on flash storage also for the data pool
3. Recurrent `MDS_CLIENT_LATE_RELEASE` warnings
    - Had some in the past, but not so insistently
    - Tracked down to subvolumes storing Redis/Grafana DBs

```
client.584712694 isn't responding to mclientcaps(revoke),
  ino 0x102d250b88f pending pAsLsXsFs issued pAsLsXsFs, sent 64.173677 sec
```

    - Vanished since upgrade to Reef

# What we learnt along the way

# Ceph Recovery Tuning

| Config Option | Default | Set to | What for |
|---|---|---|---|
| `osd_heartbeat_grace` | 20 | 300 | Avoid laggy PGs |
| `osd_delete_sleep {_hdd,_hybrid,_ssd}` | 5, 1, 1 | ~years | Save IOs on sick devices, avoiding discards |
| `osd_recovery_priority` | 5 | 250 | Prioritize backfilling wrt to client traffic |
| `osd_recovery_cost` | 20971520 | 1 | Prioritize backfilling wrt to client traffic |
| `osd_recovery_sleep {_hdd,_hybrid,_ssd}` | 0.1, 0.025, 0 | 0 | Do not wait before next recovery operation |
| `osd_max_backfills` | 1 | 50 | Avoid `backfill_wait`, then reverted to 1 |

- These were extremely helpful:
  - Initial ETA for recovery of 25% metadata: ~5 days
  - It actually took ~7 hours
- We use `wpq` scheduler; `mclock` has different set of params

# Ceph Recovery Tuning

| Config Option | Default | Set to | What for |
|---|---|---|---|
| `osd_recovery_max_active {_hdd,_ssd}` | 3, 10 | 20 | Together with `osd_max_backfills` |
| `osd_recovery_max_single_start` | 1 | 20 | Together with `osd_max_backfills` |
| `osd_op_queue_cut_off` | high | low | Send all replication ops to the strict queue |
| `osd_client_message_cap` | 256 | 1024 | Increase number of in-flight client requests |

- These were not-so-helpful:
  - No visible improvements in recovery
  - Sick SSDs already saturated by recovery ops?

- Also, some OSDs were restarted (it took 15 minutes each) to unstuck PGs not making any progress in recovery:
  - Feeling of *stuck state machine*, not fully understood

# Risk of Metadata Corruption

- Reminder: During the incident, we rebooted one node of a test cluster and it did not come back
  - Corrupted `initramfs`, `fsck` on `/` also reported issues

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- Did we copy corrupted metadata out of sick SSDs?
- Ceph is *very* careful with `O_DIRECT`, `O_SYNC`, `FLUSH`
- No inconsistencies reported while moving metadata.
  - For replica pools, this is ~= scrubbing (?)
- We deep-scrubbed metadata PGs: All `active+clean`
- `bluestore fsck` reported mismatches stats

  (but so it does on healthy clusters…)

# About Sick SSDs

- Typical random write (4k) performance: ~28k IOps
- Measured random write IOps of faulty devices: 30 IOps

- Sick devices become slow **only** when performing writes:
  - Also read operations are affected
  - Latency up to ~3s per op, and device lock-up for ~5s
- If reading only, they provide the expected performance.

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- Was it actually `Power_On_Hours` crossing 65'535?
  - Not exactly, but good enough indicator
  - Manufacturer confirmed issue with an internal counter, strongly correlated with `Power_On_Hours`

# Summary

# Thanks and Credits

- To the Ceph firefighters: Jan, Radomir, Roberto
- To our colleagues from Hardware Procurement, who confirmed the root cause with the manufacturer
- To our colleagues from Cloud, who saved hundreds of running VMs from about-to-become-sick SSDs
- To our external supporters at Clyso GmbH for switfly picking up our call for help and supporting us to HEALTH_OK

# Thank You for Listening!

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# Extras

# Digression on Probability Calculus

- Is it expected to easily move 75% of metadata, while the remaining 25% is a pain? **Yes.**
- Context:
  - 4 nodes, each in one rack
  - 3 nodes have sick SSDs, 1 node is healthy
  - CRUSH failure domain is rack
  - Data is balanced according to MGR balancer (upmap)

```
P(at least one healthy replica) is…
→ Flip it over: P(at least one healthy replica) = 1−P(all replicas sick)

→ Ways to pick 3 sick replicas out of 3: (3 3) = 1
→ Ways to pick 3 replicas out of 4 placement possible: (4 3) = 4
→ So P(all rapicas sick) is P(3 sick) / P(3 out of 4) = 1/4

→ P(at least one healthy replica): 1 - P(3 sick) = 1-1/4 = 3/4 → 75%
```

# Distribution of Power_On_Hours

